TRIGONOMETRY

Right Triangle Formulas

Angle of Measurements

- sign convention
  - o CCW (+)
  - o CW (-)
- 1 Revolution =
  - o 360° (sexagesimal)
  - o 2π radians / rad (circular)
  - o 400 gradians /grad (centisimal)
  - o 6,400 milliradians/mils (military)
- angles
  - o explementary/conjugate – 2 angles whose sum is 360°
  - o reflection angle - 180°<θ<360°

Pythagorean Relations

- $sin^2\theta+cos^2\theta=1$
- $1+tan^2\theta=sec^2\theta$
- $1+cot^2\theta=csc^2\theta$

Identities

- co-function

$\sin(90°-\theta)=\cos\theta$

$\cos(90°-\theta)=\sin\theta$

$\tan(90°-\theta)=\cot\theta$

Oblique Triangle Formulas

- sine law

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$$

$=diameter_{\text{circumscribing circle}}$

- cosine law

$a2=b^2+c^2-2bc\cos A$

$b2=a^2+c^2-2ac\cos B$

$c2=a^2+b^2-2ab\cos C$

Other Trigionometric Functions

NAME	ABBREV.	VALUE
Versed sine, versine	versin θ vers θ	$1-\cos\theta$
Versed cosine, vercosine	vercosin θ vercos θ	$1+\cos\theta$
Coversed sine, coversine	coversin θ covers θ	$1-\sin\theta$
Coversed cosine, covercosine	covercosin θ covercos θ	$1+\sin\theta$
Half versed sine, haversine	haversin θ hav θ	$\frac{1-\cos\theta}{2}$
Half versed cosine, havercosine	havercosin θ havercos θ	$\frac{1+\cos\theta}{2}$
Half coversed sine, hacoversine,	hacoversine θ hacovers θ	$\frac{1-\sin\theta}{2}$
Half coversed cosine, hacovercosine,	hacovercosin θ hacovercos θ	$\frac{1+\sin\theta}{2}$
Exterior secant, exsecant	exsec θ	$\sec\theta-1$
Exterior cosecant, excosecant	excosec θ excsc θ	$\csc\theta-1$
Chord	crd θ	$2\sin{}^{\theta}/_{2}$

- sum of 2 Angles

$\sin(A+B)=\sin A\cos B+\cos A\sin B$

$\cos(A+B)=\cos A\cos B-\sin A\sin B$

$$\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}$$

- difference of 2 angles

$\sin(A-B)=\sin A\cos B-\cos A\sin B$

$\cos(A-B)=\cos A\cos B+\sin A\sin B$

$$\tan(A-B)=\frac{\tan A-\tan B}{1+\tan A\tan B}$$

- double angle

$\sin 2A=2\sin A\cos A$

$\cos 2A = cos^2A - sin^2A = 1 - 2sin^2 A$

$$\tan 2A=\frac{2\tan A}{1-tan^2A}$$

- half angle

$$\sin\frac{A}{2}=\pm\sqrt{\frac{1-\cos A}{2}}$$

$$\cos\frac{A}{2}=\pm\sqrt{\frac{1+\cos A}{2}}$$

$$\tan\frac{A}{2}=\pm\sqrt{\frac{1-\cos A}{1+\cos A}}$$

- squares

$$sin^2A=\frac{1-\cos 2A}{2}$$

$$cos^2A=\frac{1+\cos 2A}{2}$$

$$tan^2A=\frac{1-\cos 2A}{1+\cos 2A}$$

Area of Triangle

- given 2 sides and included angle

$$A=\frac{1}{2}ab\sin C$$

- given 3 angles and 1 side

$$A=\frac{a2\sin B\sin C}{\sin A}$$

Triangles

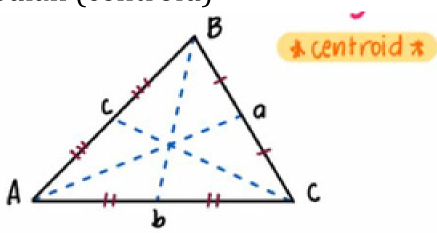
Period, Amplitude, & Frequency

FUNCTION	Period	Amplitude
$y = A\sin[B(x+C)] + D$	$2\pi/B$	A
$y = A\cos[B(x+C)] + D$	$2\pi/B$	A
$y = A\tan[B(x+C)] + D$	π/B	-

$$T = \frac{2\pi r}{v}$$

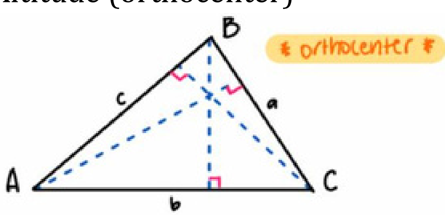
- phase shift = -C / B

Median (centroid)



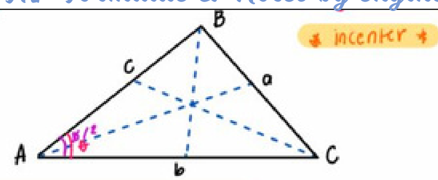
$$m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$$
$$m_b = \frac{1}{2}\sqrt{2a^2 + 2c^2 - b^2}$$
$$m_c = \frac{1}{2}\sqrt{2a^2 + 2b^2 - c^2}$$

Altitude (orthocenter)



$$h_a = \frac{2A_t}{a}$$
$$h_b = \frac{2A_t}{b}$$
$$h_c = \frac{2A_t}{c}$$

Angle Bisectors (incenter)



$$b_a = \frac{2}{b+c}\sqrt{bcs(s-b)}$$
$$b_b = \frac{2}{a+c}\sqrt{acs(s-b)}$$
$$b_c = \frac{2}{a+b}\sqrt{abs(s-c)}$$

Note:

✧ perpendicular bisectors = circumcenter

Spherical Triangle

Spherical Excess

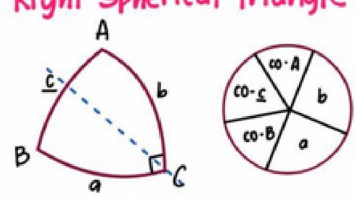
$$E = A + B + C - 180°$$

Spherical Defect

$$D = 360° - (a + b + c)$$

- greater side has greater angle
- sum of any two sides is greater than the third side
- sum of sides is less than 360°
- sum of the angles is greater than 180° and less than 540°

Right Spherical Triangle



$$\sin \text{co-}A = \cos A$$
$$\cos \text{co-}A = \sin A$$
$$\tan \text{co-}A = \cot A$$

Napier's Rule

SIN - Ta - Ad Rule

$$\sin a = \tan b \tan(\text{co-}B)$$

SIN - Co - Op Rule

$$\sin(\text{co-}c) = \cos a \cos b$$

Oblique Spherical Triangle

Law of Sine

$$\frac{\sin a}{\sin A} = \frac{\sin b}{\sin B} = \frac{\sin c}{\sin C}$$

Law of Cosine for Sides

$$\cos a = \cos b \cos c + \sin b \sin c \cos A$$

$$\cos b = \cos a \cos c + \sin a \sin c \cos B$$

$$\cos c = \cos a \cos b + \sin a \sin b \cos C$$

Law of Cosine for Angles

$$\cos A = -\cos B \cos C + \sin B \sin C \cos a$$

$$\cos B = -\cos A \cos C + \sin A \sin C \cos b$$

$$\cos C = -\cos A \cos B + \sin A \sin B \cos c$$

Terrestrial Sphere Problems:

1 minute of arc = 1 naut. mile

1 nautical mile = 6080 ft

= 1.1516 statute mile

1 statute mile = 5280 ft

1 knot = 1 nautical mile / hour

Area of Spherical Triangle

$$A = \frac{\pi R^2 E}{180^\circ}$$

Volume of Spherical Triangle

$$V = \frac{\pi R^3 E}{540^\circ}$$

$$\tan\frac{E}{4} = \sqrt{\tan\frac{1}{2}s \tan\frac{1}{2}(s-a) \tan\frac{1}{2}(s-b) \tan\frac{1}{2}(s-c)}$$

$$s = \frac{a+b+c}{2}$$